If all spring constants are identical i.e., $k_1 = k_2 = \ldots = k_n = k$ then

$$k_p = n\,k \qquad (10.49)$$

This implies that the effective spring constant increases by a factor n. Hence, for the springs in parallel connection, the effective spring constant is greater than individual spring constant.

Note

The spring constant is inversely proportional to the length of the spring

$$k \propto \frac{1}{length\ of\ the\ spring}$$

If the spring is cut into two pieces, one piece with length l_1 and other with length l_2, such that $l_1 = nl_2$, then spring constant of first length is $k_1 = \frac{k(n+1)}{n}$ and spring constant of second length is $k_2 = (n+1)\,k$, where k is the original spring constant before cutting into pieces.

EXAMPLE 10.10

Consider two springs with force constants 1 N m^{-1} and 2 N m^{-1} connected in parallel. Calculate the effective spring constant (k_p) and comment on k_p.

Solution

$$k_1 = 1\ \text{N m}^{-1},\ k_2 = 2\ \text{N m}^{-1}$$

$$k_p = k_1 + k_2\ \text{N m}^{-1}$$

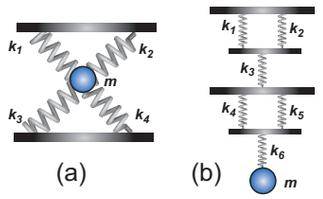
$$k_p = 1 + 2 = 3\ \text{N m}^{-1}$$

$$k_p > k_1 \text{ and } k_p > k_2$$

Therefore, the effective spring constant is greater than both k_1 and k_2.

EXAMPLE 10.11

Calculate the equivalent spring constant for the following systems and also compute if all the spring constants are equal:



Solution

a. Since k_1 and k_2 are parallel, $k_u = k_1 + k_2$
Similarly, k_3 and k_4 are parallel, therefore, $k_d = k_3 + k_4$
But k_u and k_d are in series,

therefore, $k_{eq} = \frac{k_u k_d}{k_u + k_d}$

If all the spring constants are equal then, $k_1 = k_2 = k_3 = k_4 = k$
Which means, $k_u = 2k$ and $k_d = 2k$

Hence, $k_{eq} = \frac{4k^2}{4k} = k$

b. Since k_1 and k_2 are parallel, $k_A = k_1 + k_2$
Similarly, k_4 and k_5 are parallel, therefore, $k_B = k_4 + k_5$
But k_A, k_3, k_B, and k_6 are in series,

therefore, $\frac{1}{k_{eq}} = \frac{1}{k_A} + \frac{1}{k_3} + \frac{1}{k_B} + \frac{1}{k_6}$

If all the spring constants are equal then, $k_1 = k_2 = k_3 = k_4 = k_5 = k_6 = k$
which means, $k_A = 2k$ and $k_B = 2k$

$$\frac{1}{k_{eq}} = \frac{1}{2k} + \frac{1}{k} + \frac{1}{2k} + \frac{1}{k} = \frac{3}{k}$$

$$k_{eq} = \frac{k}{3}$$

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